

Name: _____



Medical professional

Task A: Arithmetic in medical professions

1) The table shows a patient's fluid input and output records during four hours.

time	input			output			fluid balance (ml)
	drink (ml)	liquid medication (ml)	IV fluids (ml)	urine (ml)	vomit or aspiration (ml)	drain (ml)	
09:00	200	0	125	0	0	5	
10:00	0	10	125	300	0	25	
11:00	0	0	125	0	200	15	
12:00	150	0	125	0	0	10	

a) Calculate the fluid balance for each hour.

b) Calculate the overall fluid balance for the entire 4-hour period.



Task B: Using formulae in medical professions

1) A patient has a mass of 90 kg

They are prescribed 600 mg of a drug called rifampicin.

The drug comes vials which each have a volume of 10 ml.

Each vial contains 300 mg of the drug.

It need to be diluted up to 500 ml for the IV bag.

Once diluted, it needs to be administered continuously over 3 hours.

a) Use the formula below to calculate the dose required.

$$\text{drug dose (ml)} = \frac{\text{amount of drug prescribed (mg)}}{\text{amount of drug per vial (mg)}} \times \text{volume of each vial (ml)}$$

b) The 'administration rate' is the amount of the diluted drug that is given to the patient per hour from the IV bag. Calculate administration rate for this scenario.

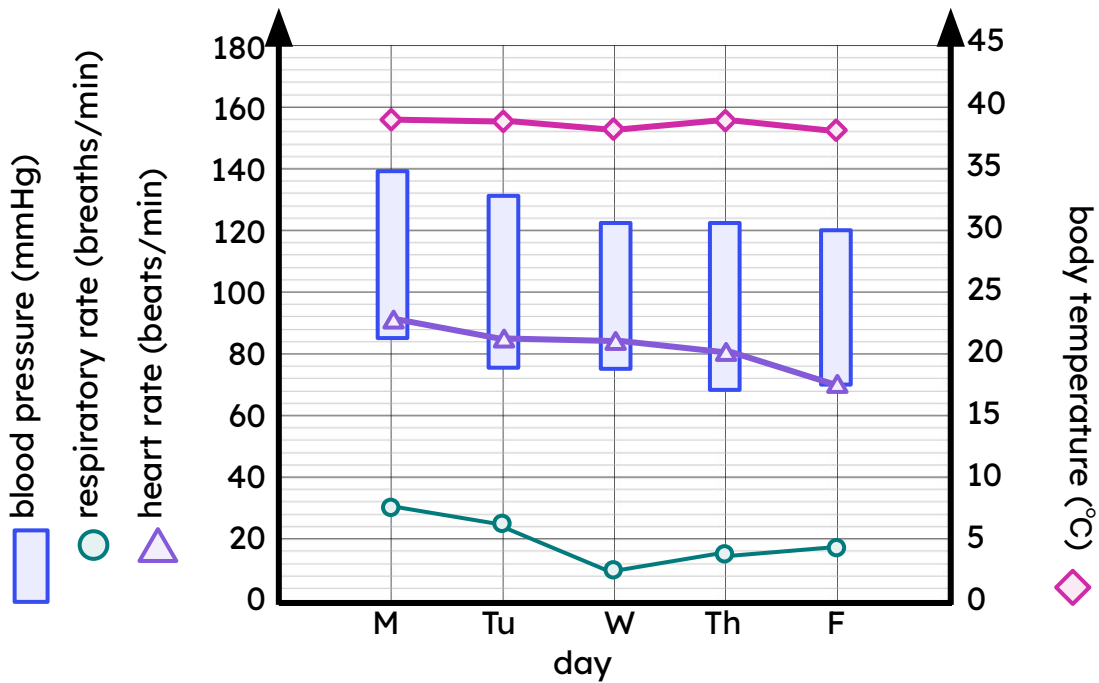
c) 1 mg = 1000 mcg. How many mcg of rifampicin was prescribed to the patient?

d) Use your answers from parts (b) to (c) and the formula below to calculate the mcg/kg/min settings for the IV drip.

$$\text{drip settings (mcg/kg/min)} = \frac{\text{administration rate (ml/hr)} \times \text{amount prescribed (mcg)}}{\text{bag volume (ml)} \times \text{patient mass (kg)} \times 60}$$

Task C: Using formulae in medical professions

1) The graph shows a patient's vital signs over five days.



- On what day was their respiratory rate at its lowest?
- What was their heart rate on Tuesday?
- Lucas thinks their body temperature was 155°C on Monday.
What mistake has Lucas made?
- What was their body temperature on Monday?
- Describe the trend in their blood pressure over the five days.
- A healthy blood pressure recording is around $\frac{120}{70}$
On what day(s) is the patient's blood pressure around these levels?